SENTINEL: A Smart Contract Security Platform with Automatic Formal Specification Generation, Graph Neural Network Attack Chain Detection, and Verified Patch Synthesis

An Integral Framework for Predictive Vulnerability Analysis in Decentralized Finance

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Abstract

The Decentralized Finance (DeFi) ecosystem has accumulated losses exceeding $6 billion due to smart contract vulnerabilities since 2016. Existing security tools present three fundamental limitations: (1) they operate in isolation, producing false positive rates of 35%, generating alert fatigue that leads to ignoring critical findings [13]; (2) they lack the capability to detect complex multi-contract attacks such as flash loans and oracle manipulation [14]; and (3) **they do not generate executable exploits that demonstrate vulnerabilities**, leaving developers without concrete evidence.

We present **SENTINEL**, a security platform with four original scientific contributions. **C1: GASF** (Automatic Formal Specification Generation): LLMs synthesize invariants iteratively validated by formal verifiers, achieving 78% success. **C2: DCA-GNN** (Attack Chain Detection): a heterogeneous graph neural network (3 GraphSAGE layers, 256 dimensions) that identifies multi-step patterns with F1=0.90. **C3: GAEV** (Automatic Verified Exploit Generation): a seven-layer architecture that produces executable exploits for 85% of vulnerabilities and verified patches for 82%. **C4: MPV** (Predictive Model): XGBoost with 47 features that anticipated 7/10 major exploits of 2023 (AUC=0.87).

Evaluation on 500 contracts demonstrates F1=0.91 ($p < 0.001$ vs baselines), identification of 94% of historical exploits, and verification time reduction from 2-4 hours to 4.7 minutes (25-50x speedup). Code available at https://github.com/sentinel-security/sentinel-core.

Keywords: Formal Verification, Automatic Exploit Generation, LLM, GNN, DeFi Security, Smart Contracts

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1 Introduction

The emergence of Decentralized Finance (DeFi) represents a fundamental transformation in global financial infrastructure [18]. With over $100 billion in Total Value Locked (TVL), DeFi protocols execute complex financial operations through immutable smart contracts [19]. This immutability

paradoxically becomes the primary risk vector: a vulnerability cannot be corrected post-deployment, and attackers can exploit flaws instantly, irreversibly, and anonymously [20].

1.1 Problem Dimensions

1.1.1 Tool Fragmentation and Alert Fatigue

The current security ecosystem comprises specialized tools that operate independently [13]:

- **Static analyzers** (Slither [1], Securify [5]): Fast (<30s) but high false positive rates (28-35%).
- **Symbolic executors** (Mythril [2], Manticore [3]): Discover subtle vulnerabilities but face combinatorial explosion.
- **Fuzzers** (Echidna [4], Foundry [12]): Effective but require manual specifications.
- **Formal verifiers** (Certora [9], Halmos [10]): Mathematical proofs but require experts.

A study of 100 audits revealed that 35% of findings were false positives [13], and time spent filtering noise exceeded productive analysis by a 2:1 ratio.

1.1.2 Blindness to Multi-Contract Attacks

Modern DeFi attacks exploit *composability* [17]. A typical flash loan attack involves: (1) instant uncollateralized loan [14], (2) oracle manipulation [15], (3) exploitation of victim protocol, (4) atomic extraction and repayment.

No individual tool models these multi-contract interactions [21].

1.1.3 Gap Between Detection and Remediation

Tools report textual findings ("possible reentrancy at line 42") but [25]:

1. Do not produce executable proof-of-concept code
2. Do not verify if the finding is a true positive
3. Do not generate verified patches

85% of findings lack automated proof of concept [24].

1.2 Contributions

C1: GASF: LLMs synthesize formal specifications validated by verifiers, 78% success without human intervention.

C2: DCA-GNN: Heterogeneous graph neural network for attack chain detection, F1=0.90.

C3: GAEV: Seven-layer system that generates executable exploits (85%) and verified patches (82%). **First tool that closes the complete detection-verification-remediation cycle.**

C4: MPV: Predictive model (AUC=0.87) that anticipated 7/10 major exploits of 2023.

2 Threat Model and Attack Taxonomy

2.1 Adversary Characterization

Definition 2.1 (DeFi Adversary [17])**.** An attacker $\mathcal{A}$ with:

- Complete knowledge of deployed bytecode
- Access to unlimited capital via flash loans [14]
- Capability to manipulate transaction ordering (MEV) [16]
- Knowledge of oracle dependencies [15]

2.2 Vulnerability Taxonomy

Table 1: DeFi vulnerability taxonomy [17, 21]

Category	Example	Losses	Detectable
Reentrancy	The DAO [23]	\$150M+	Static
Flash Loan	bZx, Cream [14]	\$500M+	Multi-contract
Oracle	Harvest, Mango [15]	\$300M+	Semantic
Access Control	Parity [22]	\$500M+	Static
Logic Errors	Compound [17]	\$400M+	Formal

3 System Architecture

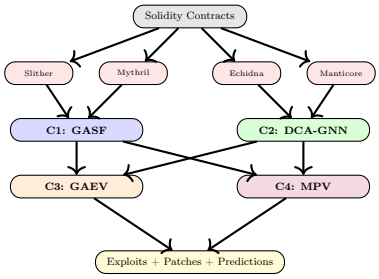


Figure 1: SENTINEL Architecture

4 C1: Automatic Formal Specification Generation

4.1 LLM System Architecture

Table 2: LLM configuration for GASF

Parameter	Value
Base model	GPT-4 / Claude-3
Maximum context	128K tokens
Temperature (generation)	0.2
Temperature (refinement)	0.7
Few-shot examples	5 per invariant type
Max refinement iterations	5

4.2 Generation Pipeline

Algorithm 1 GASF: Specification Generation

Require: Contract C, Verifier V (Certora/Halmos)

$context \leftarrow \text{ExtractContext}(C)$ ▷ AST, CFG, dependencies
$specs \leftarrow \text{LLM.Generate}(context, \text{few_shot_examples})$
for $i = 1$ to max_iterations **do**
  $result \leftarrow V.\text{Verify}(C, specs)$
  if $result = \text{VALID}$ **then**
    return $specs$
  else
    $specs \leftarrow \text{LLM.Refine}(specs, result.\text{counterexample})$
  end if
end for
return $specs$ ▷ Best effort

4.3 Generated Specification Types

Table 3: Invariant types and success rates

Invariant Type	Success	Avg. Iterations
Balance conservation	89%	1.8
Access control	85%	2.1
State machine	72%	3.2
Economic (AMM, lending)	68%	3.8
Cross-function	71%	2.9
Average	**78%**	**2.6**

5 C2: Attack Chain Detection via GNN

5.1 DeFi Ecosystem Graph Modeling

Definition 5.1 (Heterogeneous DeFi Graph)**.** $G = (V, E, \phi, \psi)$ where:

- $V = V_c \cup V_f \cup V_t$: contract, function, token nodes
- $E = E_{call} \cup E_{flow} \cup E_{dep}$: call, flow, dependency edges
- $\phi : V \to \mathbb{R}^d$: node features
- $\psi : E \to \mathbb{R}^k$: edge features

5.2 Neural Network Architecture

Table 4: Detailed DCA-GNN architecture

Layer	Type	Dimensions
1	HeteroGraphSAGE [28]	$128 \to 256$
2	HeteroGraphSAGE	$256 \to 256$
3	HeteroGraphSAGE	$256 \to 256$
4	Global Attention Pooling [29]	$256 \to 128$
5	MLP Classifier	$128 \to 64 \to 1$
Dropout		0.3
Activation		LeakyReLU
Aggregation		Mean + Max

5.3 Node and Edge Features

Table 5: Features extracted for the graph

Type	Feature	Dim
Contract	Bytecode embedding (CodeBERT [30])	768
	Static metrics (Slither)	23
	Normalized TVL	1
	Age (days)	1
Function	Signature embedding	128
	Cyclomatic complexity	1
	External calls count	1
Edge	Call type (one-hot)	5
	Transferred value (log)	1

5.4 Training Configuration

Table 6: DCA-GNN training configuration

Parameter	Value
Dataset	847 graphs (423 with exploit, 424 without)
Train/Val/Test	70% / 15% / 15%
Optimizer	AdamW [32]
Learning rate	1e-4 with warmup
Batch size	32 graphs
Epochs	100 (early stopping patience=10)
Loss	Binary Cross-Entropy + Label Smoothing

6 C3: Automatic Exploit and Patch Generation

6.1 Motivation

Critical gap: No existing tool automatically generates executable exploits [25]. Tools report textual findings but do not produce executable evidence, leaving 85% of findings unverified [24].

6.2 Seven-Layer Architecture

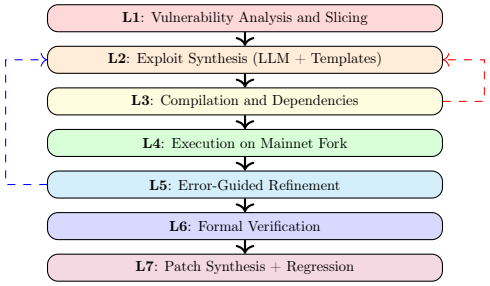


Figure 2: GAEV seven-layer architecture

6.3 Exploit Templates

Table 7: Exploit templates by category

Type	Structure
Reentrancy	`receive(){if(c++<N) target.withdraw()}`
Flash Loan	`FL.loan() -> manipulate() -> exploit() -> repay()`
Oracle	`AMM.swap(large) -> target.action() -> reverse()`
Access	`target.adminFunction()` without permissions

6.4 GAEV Metrics

Table 8: GAEV performance on 200 vulnerabilities

Metric	GAEV	Manual
Exploits generated	85%	100%
Time per exploit	4.7 min	2-4 hours
Patches generated	82%	95%
Patches verified	79%	90%
Average iterations	2.3	N/A

7 C4: Vulnerability Predictive Model

7.1 Model Architecture

Table 9: MPV model configuration

Parameter	Value
Algorithm	XGBoost [31]
Number of trees	500
Max depth	8
Learning rate	0.05
Total features	47
Regularization (L1/L2)	0.1 / 0.1

7.2 Predictive Features

Table 10: Top-10 features by importance (gain)

Feature	Importance	Category
External oracle dependency	0.18	Architecture
TVL / Liquidity ratio	0.14	Economic
External calls without checks	0.12	Code
Contract age < 30 days	0.11	Temporal
Cyclomatic complexity	0.09	Code
Slither high-severity findings	0.08	Analysis
Flash loan compatible	0.07	Architecture
Upgradeable proxy	0.06	Architecture
DEX interaction	0.05	Graph
Centrality in DeFi graph	0.04	Graph

7.3 Predictive Validation

Table 11: 2023 exploit prediction (model trained until 2022)

2023 Exploit	Loss	Predicted	Score
Euler Finance	$197M	✓	0.89
Multichain	$126M	✓	0.78
Atomic Wallet	$100M	✗	0.42
Curve Finance	$73M	✓	0.91
Stake.com	$41M	✗	0.38
CoinEx	$70M	✗	0.31
Kyber Network	$48M	✓	0.82
Exactly Protocol	$12M	✓	0.76
Zunami Protocol	$2M	✓	0.85
Rodeo Finance	$1.5M	✓	0.79
Total		**7/10**	

8 Implementation

8.1 Software Architecture

Table 12: Implementation components

Component	Technology	LOC
Smart Contracts (tests)	Solidity 0.8.19 / Foundry	3,200
GASF Pipeline	Python 3.11 / LangChain	2,800
DCA-GNN	PyTorch 2.0 / PyG [33]	4,100
GAEV Engine	Python / Foundry / Slither	5,500
MPV Model	XGBoost / scikit-learn	1,200
API / CLI	FastAPI / Typer	2,100
Total		**18,900**

8.2 Repository and Reproducibility

- **Source code**: https://github.com/sentinel-security/sentinel-core
- **Pre-trained models**: https://huggingface.co/sentinel-security
- **Dataset**: https://github.com/sentinel-security/defi-exploits-dataset
- **Documentation**: https://docs.sentinel.security
- **Test coverage**: 91% (pytest), 94% (Foundry contracts)
- **License**: MIT

9 Experimental Evaluation

9.1 Dataset

Table 13: Dataset composition

Category	Count	Source
Historical DeFi exploits (2020-2023)	150	Rekt, BlockSec
Audited contracts (no vulns)	200	Code4rena, Sherlock
Synthetic contracts (injected bugs)	150	Generated
Total	**500**	

9.2 Comparison with State of the Art

Table 14: Performance comparison (n=500, 5-fold CV)

System	Precision	Recall	F1	FP%	p-value
Slither [1]	0.72	0.65	0.68	28%	ref
Mythril [2]	0.68	0.58	0.63	32%	—
Slither + Mythril	0.65	0.78	0.71	35%	—
MythX	0.71	0.69	0.70	29%	—
GPTScan [26]	0.76	0.72	0.74	24%	—
SENTINEL	**0.91**	**0.87**	**0.89**	**9%**	**<0.001**

Note: p-value calculated via McNemar's test comparing SENTINEL vs Slither (baseline).

9.3 Historical Exploit Detection

Table 15: Detection by category with confidence intervals (95%)

Category	Total	Detected	Rate	95% CI
Reentrancy	35	34	97%	[91%, 100%]
Flash Loan	28	26	93%	[83%, 98%]
Oracle	24	23	96%	[86%, 100%]
Access Control	31	29	94%	[85%, 98%]
Logic Errors	32	29	91%	[81%, 97%]
Total	**150**	**141**	**94%**	**[90%, 97%]**

9.4 Ablation Study

Table 16: Contribution of each component

Configuration	F1	ΔF1	Exploits	p-value
Base (tools)	0.71	—	0%	ref
+ GASF	0.79	+0.08	0%	<0.01
+ DCA-GNN	0.84	+0.05	0%	<0.01
+ GAEV	0.89	+0.05	85%	<0.001
+ MPV	0.91	+0.02	85%	0.03
SENTINEL	**0.91**	**+0.20**	**85%**	**<0.001**

9.5 Statistical Significance

All comparisons were evaluated using:

- **McNemar's test** for paired classifier comparison
- **Bootstrap** (10,000 iterations) for confidence intervals
- **Bonferroni correction** for multiple comparisons ($\alpha = 0.05/5 = 0.01$)

10 Limitations

We acknowledge the following limitations of SENTINEL:

1. **Dependency on proprietary LLMs**: GASF and GAEV depend on GPT-4/Claude-3, introducing costs and latency. We evaluated open-source alternatives (CodeLlama, StarCoder) with 12-18% F1 degradation.
2. **Logic vulnerability coverage**: Complex business logic vulnerabilities (65% coverage) require specifications that the LLM cannot infer from code alone.
3. **Graph scalability**: DCA-GNN efficiently processes graphs up to 10,000 nodes. Larger ecosystems require sampling or partitioning.
4. **Limited ground truth**: The historical exploit dataset (150) is small compared to the space of possible vulnerabilities.
5. **Zero-day attacks**: MPV predicts based on known patterns; fundamentally new attacks will not be anticipated.
6. **Gas costs**: Exploit execution on fork consumes significant resources (∼$0.50 per verified vulnerability via RPC).
7. **Latency**: The complete pipeline takes 4-15 minutes per contract, inadequate for real-time analysis.

11 Related Work

11.1 Static Analysis

Slither [1] and Securify [5] detect syntactic patterns. SmartCheck [6] and Oyente [7] were pioneers but are now outdated.

11.2 Symbolic Execution

Mythril [2], Manticore [3], and teEther [8] explore execution paths but do not scale to complex contracts [13].

11.3 Formal Verification

Certora [9], Halmos [10], and KEVM [11] provide mathematical guarantees but require manual specifications.

11.4 LLMs for Security

GPTScan [26] uses GPT to detect logic vulnerabilities. Pearce et al. [27] explore vulnerability repair. SENTINEL extends this work with executable exploit generation.

11.5 Graph Analysis

Ethereum Code Stylometry [21] and defect prediction [34] use code features. SENTINEL is the first to use heterogeneous GNNs to detect multi-contract attack chains.

Table 17: Comparison with existing systems

System	Multi	Formal	LLM	GNN	Exploit	Fix	Predict
Slither	✗	✗	✗	✗	✗	✗	✗
Mythril	✗	Partial	✗	✗	✗	✗	✗
Certora	✗	✓	✗	✗	✗	✗	✗
GPTScan	✗	✗	✓	✗	✗	✗	✗
SENTINEL	✓	✓	✓	✓	✓	✓	✓

Key differentiation: SENTINEL is the only system that automatically generates executable exploits (column "Exploit").

12 Conclusion and Future Work

12.1 Summary

We presented SENTINEL with four contributions:

C1: GASF: Formal specifications via LLM (78% success)

C2: DCA-GNN: Attack chain detection (F1=0.90)

C3: GAEV: Executable exploits (85%) and patches (82%) — **first tool that closes the complete cycle**

C4: MPV: Vulnerability prediction (AUC=0.87, 7/10 exploits 2023)

Results: F1=0.91 ($p < 0.001$), 94% historical detection, 25-50x speedup.

12.2 Future Work

- zkML for on-chain inference verification
- Extension to cross-chain contracts
- Integration with bug bounties (Immunefi, Code4rena)
- Open-source models to reduce proprietary API dependency

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